HW3 due by Wednesday Feb. 25 by 11:59 PM

The create_expense.sql script (you can find it on Blackboard in the Homework 3 folder) creates a database which contains the 7 tables described below. The data contains a tracking system for expense reports filed by employees at a manufacturing company. **Please watch the Panopto "video 5 for HW3" before doing this Homework 3.**

If you use Chat GPT, please use the "Share" button (looks like 'upward arrow') in the right corner of ChatGPT chat, and 'copy link' and share the link to that chat in this Word document and briefly explain how you used it for your Homework (for each HW question if you used it). No points off will be taken for using ChatGPT (it is allowed to use it for Homework) but you are required to share the link to a chat if you used it.

More information on how to share a chat here: https://help.openai.com/en/articles/7925741-chatgpt-shared-links-faq

employees	**Field Description**
Ssn (pk)	Unique SSN ID# for employee
First_name	Employee first name
Last_name	Employee last_name
Dept (fk)	Dept ID#
Start_year	Year of employment

trips	**Field Description**
Employee (pk, fk)	SSN of employee travelling
Trip_ID (pk)	Unique Trip ID#
Start_date	Start date of trip
End_date	End date of trip
Reason_code (fk)	Code for reason for trip

expenses	**Field Description**
Employee (pk, fk)	SSN of employee travelling
Trip_id (pk, fk)	Unique Trip ID#
Expense_seq (pk)	Sequence# for expense report line item
Account_no (fk)	Account number for line item
Gross_amount	Gross dollar amount of line item
tax	Sales tax (if applicable) of line item

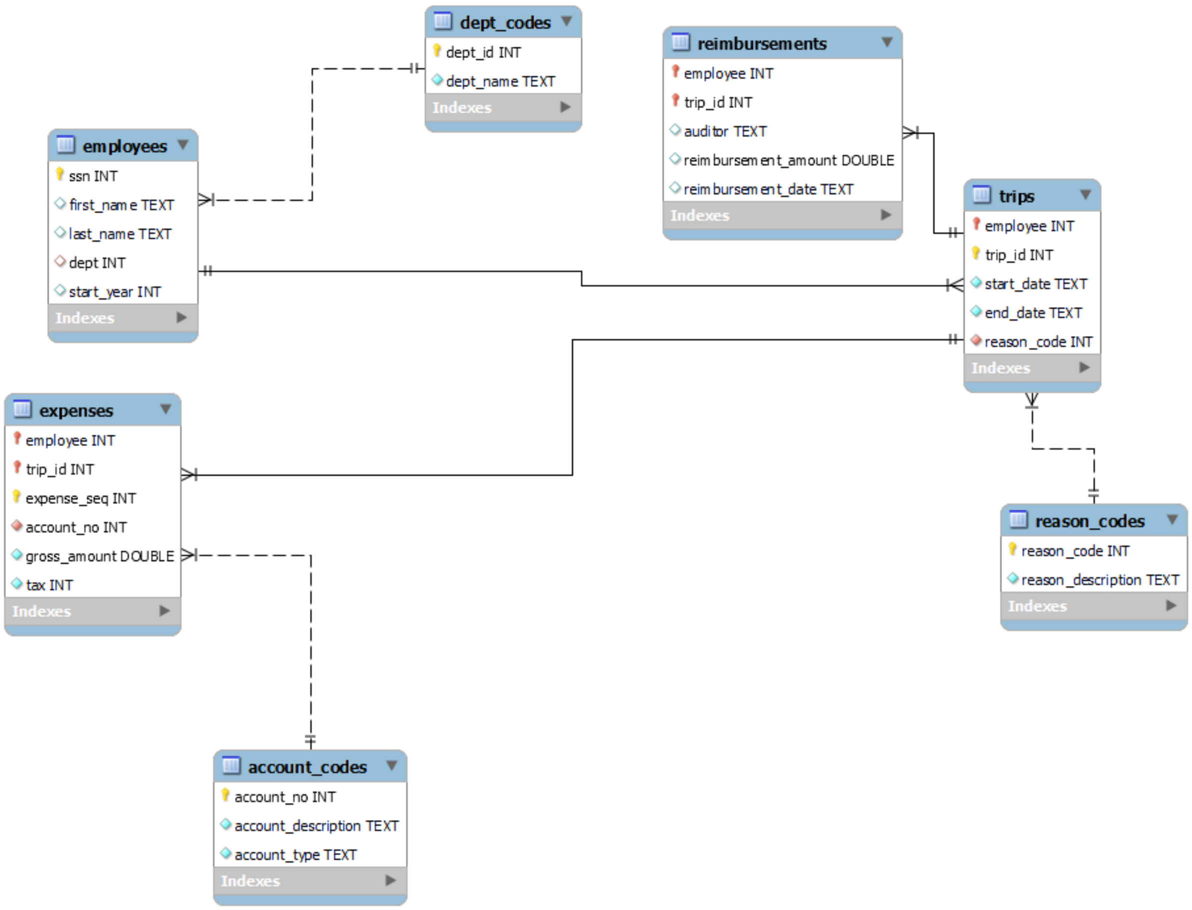
dept_codes	**Field Description**

Dept_ID (pk)	Dept ID#
Dept_name	Name of department

Reason_codes	**Field Description**
Reason_code (pk)	Reason ID#
Reason_description	Description of reason for trip

account_codes	**Field Description**
Account_no (pk)	Account ID#
Account_description	Description of account
Account_type	Category of account

reimbursements	**Field Description**
Employee (pk, fk)	SSN of employee travelling
Trip_id (pk, fk)	Unique Trip ID#
Auditor	Auditor last name
Reimbursement_amount	Amount of reimbursement
Reimbursement_date	Date of reimbursement



Please put all of your work into **this single Word doc**.

1. (60 points) Design and create a data warehouse for the Expense database. The decisions about which fields to include and how to aggregate the data are left to you. You do not need to include every single data point from the 7 tables given. Use your judgement as to what will be interesting/useful for the organization. But please make sure that you pull (combine) data from **at least four tables** and compute relevant aggregate statistics. Please see many examples from class lectures and you may adapt those codes for your purpose (for this dataset).

Submit a screenshot of the first 25 rows of your data warehouse (paste into this Word document) and the SQL code that you used to create it. Please copy and paste your SQL code into this Word document.

USE expense;

DROP TABLE IF EXISTS Expense_DW;

```
CREATE TABLE Expense_DW AS
SELECT
   -- Dimension Attributes
   t.Employee,
   t.Trip_ID,
   e.First_name,
   e.Last_name,
   d.Dept_name,
   r.Reason_description,
   t.Start_date,
   t.End_date,

   -- Fact Measures
   COUNT(ex.Expense_seq) AS Total_Line_Items,
   SUM(ex.Gross_amount) AS Total_Gross_Amount,
   SUM(COALESCE(ex.tax,0)) AS Total_Tax_Amount,
   SUM(ex.Gross_amount + COALESCE(ex.tax,0)) AS Total_Net_Amount

FROM trips t
JOIN employees e
   ON e.Ssn = t.Employee
JOIN dept_codes d
   ON d.Dept_ID = e.Dept
JOIN reason_codes r
   ON r.Reason_code = t.Reason_code
JOIN expenses ex
   ON ex.Employee = t.Employee
  AND ex.Trip_id = t.Trip_ID

GROUP BY
   t.Employee,
   t.Trip_ID,
   e.First_name,
   e.Last_name,
   d.Dept_name,
   r.Reason_description,
   t.Start_date,
   t.End_date;

   SHOW TABLES;
```

create_expense | expense_queries_to_start_HW3*

Limit to 1000 rows

```
SHOW TABLES;


SELECT
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content: | Fetch rows:

Employee	Trip_ID	First_name	Last_name	Dept_name	Reason_description	Start_date	End_date	Total_Line_Items	Total_Gross_Amount	Total_Tax_Amount	Total_Net_Amount
152919947	3795	Elly	Duggan	Finance	conference	3/13/2017	3/15/2017	6	2695.73	180	2875.73
152919947	3955	Elly	Duggan	Finance	supplier visit	10/24/2017	10/26/2017	4	673.03	103	776.03
152919947	4639	Elly	Duggan	Finance	conference	10/23/2017	10/25/2017	5	1125.38	127	1252.38
152919947	6044	Elly	Duggan	Finance	sales call	2/12/2017	2/16/2017	10	1168.69	103	1271.69
152919947	6218	Elly	Duggan	Finance	site visit	7/11/2017	7/14/2017	4	933.54	148	1081.54
152919947	6310	Elly	Duggan	Finance	off-site meeting	11/16/2017	11/17/2017	2	1322.21	175	1497.21
152919947	6463	Elly	Duggan	Finance	conference	8/4/2017	8/5/2017	2	1024.07	0	1024.07
270009780	3616	Teodor	Lynch	Finance	supplier visit	11/21/2017	11/23/2017	3	757.16	122	879.16
270009780	3979	Teodor	Lynch	Finance	site visit	1/18/2017	1/21/2017	5	1947.6	275	2222.6
270009780	4300	Teodor	Lynch	Finance	sales call	4/26/2017	4/29/2017	9	1951.6999999999998	257	2208.7
270009780	5704	Teodor	Lynch	Finance	supplier visit	12/3/2017	12/4/2017	3	734.1	116	850.1
270009780	6838	Teodor	Lynch	Finance	supplier visit	9/13/2017	9/16/2017	10	1134.87	128	1262.87
333696062	2730	Frank	Skinner	Finance	sales call	10/3/2017	10/4/2017	1	602.11	100	702.11
333696062	2941	Frank	Skinner	Finance	sales call	5/31/2017	6/2/2017	6	492.40999999999997	53	545.41
333696062	3716	Frank	Skinner	Finance	sales call	7/1/2017	7/4/2017	10	1178.05	136	1314.05
370776732	6348	Kyron	Pham	Finance	off-site meeting	9/18/2017	9/20/2017	6	1200.44	123	1323.44
457581906	6133	Dania	Chen	Finance	site visit	6/30/2017	7/3/2017	11	1672.56	189	1861.56
484894163	2941	Javan	Everett	Finance	sales call	5/31/2017	6/2/2017	1	23.7	0	23.7
484894163	3390	Javan	Everett	Finance	supplier visit	7/25/2017	7/27/2017	5	763.04	112	875.04
484894163	3972	Javan	Everett	Finance	off-site meeting	12/7/2017	12/8/2017	2	671.42	111	782.42
721480714	3513	Sheldon	Drummond	Finance	off-site meeting	2/10/2017	2/13/2017	10	1399.74	168	1567.74
721480714	4170	Sheldon	Drummond	Finance	off-site meeting	3/28/2017	3/31/2017	11	2087.29	162	2249.29
721480714	5271	Sheldon	Drummond	Finance	off-site meeting	5/30/2017	6/2/2017	14	1585.0200000000002	176	1761.020000000...
721480714	5628	Sheldon	Drummond	Finance	conference	10/31/2017	11/1/2017	2	430.53000000000003	0	430.5300000000...
880208202	3039	Marguerite	Alcock	Finance	conference	1/12/2017	1/14/2017	6	1712.94	100	1812.94

2. (40 points) Create **four** SQL queries on your data warehouse that answer interesting important questions. At least two queries should be more than simple queries. For example, more complex queries could include Joins, a Group By element or a subquery or use some aggregate functions and summary calculations (see examples in the class lectures' slides).

Submit a copy of each query SQL code (paste into this Word document), and the screenshot of each query results (or the first 25 rows if it is longer) and a one or two sentence description of the question your SQL code was addressing and what you found in the results.

Query #1 - This query summarizes total and average trip spending by department. It helps identify which departments contribute the most to overall travel expenses. The three departments that contribute the most to travel expenses are Sales, Purchasing and Quality. While these departments account for the most total spending, the most highest average spending per trip are the following departments Executive, Human Resources and Logistics.

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

Dept_name	Total_Trips	Total_Department_Spending	Avg_Spending_Per_Trip
Sales	379	478344.99	1262.1239841688655
Purchasing	167	227742.42999999996	1363.7271257485029
Quality	106	212834.54000000004	2007.8730188679249
Executive	47	197212.05999999994	4196.001276595744
Logistics	57	123268.99999999997	2162.614035087719
Human Resources	33	87814.80999999998	2661.0548484848478
Operations	33	59187.37999999999	1793.5569696969694
Finance	33	38974.32000000001	1181.0400000000002
Information Technology	14	19375.600000000002	1383.9714285714288

Query #2 - This query summarizes total and average trip spending by department. It helps identify which departments contribute the most to overall travel expenses. The conclusion is that conferences by have the highest total spending and avergae Trip cost.

```
SELECT
  Reason_description,
  COUNT(*) AS Trip_Count,
  AVG(Total_Net_Amount) AS Avg_Trip_Cost,
  SUM(Total_Net_Amount) AS Total_Spending
FROM Expense_DW
GROUP BY Reason_description
HAVING COUNT(*) >= 5
ORDER BY Avg_Trip_Cost DESC;
```

Reason_description	Trip_Count	Avg_Trip_Cost	Total_Spending
conference	172	2454.0259302325585	422092.4600000001
off-site meeting	104	2238.2228846153853	232775.18000000005
site visit	168	1542.855595238095	259199.73999999996
supplier visit	152	1318.8845394736836	200470.4499999999
sales call	273	1209.5871794871796	330217.30000000005

Query #3 – This query identifies trips with costs above the overall average. It helps flag unusually expensive trips for review or policy evaluation. The conclusion is that the most expensive trips are taken by executive for off-site meetings or conferences. A potential solution might be for off-site meetings and conferences to be attended virtually when possible.

```
SELECT *
FROM Expense_DW
WHERE Total_Net_Amount >
(
   SELECT AVG(Total_Net_Amount)
   FROM Expense_DW
)
ORDER BY Total_Net_Amount DESC;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

Employee	Trip_ID	First_name	Last_name	Dept_name	Reason_description	Start_date	End_date	Total_Line_Items	Total_Gross_Amount	Total_Tax_Amount	Total_Net_Amount
477501922	4530	Amara	Graham	Executive	off-site meeting	8/11/2017	8/16/2017	27	6858.41	572	7430.41
557696451	4279	Iqra	Mcnally	Executive	site visit	7/8/2017	7/15/2017	31	5825.05	699	6524.05
557696451	5264	Iqra	Mcnally	Executive	off-site meeting	11/2/2017	11/6/2017	20	5913.28	498	6411.28
180604167	2945	Macaulay	Delaney	Executive	off-site meeting	7/11/2017	7/16/2017	15	5803.54	460	6263.54
247505332	5102	Morris	Chavez	Executive	off-site meeting	6/11/2017	6/16/2017	25	5683.07	556	6239.07
557696451	5845	Iqra	Mcnally	Executive	conference	7/11/2017	7/17/2017	16	5506.2300000000005	580	6086.2300000000005
180604167	6894	Macaulay	Delaney	Executive	conference	5/13/2017	5/18/2017	25	5536.52	457	5993.5199999999995
477501922	3623	Amara	Graham	Executive	sales call	7/13/2017	7/20/2017	40	5222.45	557	5779.45
180604167	4767	Macaulay	Delaney	Executive	conference	1/15/2017	1/19/2017	15	5235.419999999999	489	5724.419999999999
995619281	6659	Duke	Witt	Logistics	off-site meeting	9/18/2017	9/23/2017	18	5212.38	389	5601.38
557696451	6798	Iqra	Mcnally	Executive	off-site meeting	12/9/2017	12/14/2017	21	4889.370000000001	504	5393.370000000001
557696451	2989	Iqra	Mcnally	Executive	conference	4/25/2017	4/30/2017	12	4903.57	405	5308.57
303083671	6293	Kit	Dunn	Executive	off-site meeting	5/29/2017	6/4/2017	18	4895.139999999999	325	5220.139999999999
477501922	4404	Amara	Graham	Executive	conference	11/11/2017	11/15/2017	14	4703.11	428	5131.11
180604167	6856	Macaulay	Delaney	Executive	supplier visit	6/15/2017	6/20/2017	24	4612.88	504	5116.88
303083671	6949	Kit	Dunn	Executive	off-site meeting	2/17/2017	2/21/2017	15	4755.64	298	5053.64
303083671	6211	Kit	Dunn	Executive	site visit	3/21/2017	3/27/2017	16	4253.79	504	4757.79
557696451	7089	Iqra	Mcnally	Executive	conference	2/16/2017	2/21/2017	22	4340.99	414	4754.99
247505332	4619	Morris	Chavez	Executive	site visit	9/16/2017	9/22/2017	13	4224.06	524	4748.0599999999995
180604167	5389	Macaulay	Delaney	Executive	conference	7/4/2017	7/9/2017	21	4334.67	305	4639.67
802792878	6954	Mabel	Jimenez	Logistics	conference	9/15/2017	9/20/2017	24	4339.96	283	4622.96
247505332	3021	Morris	Chavez	Executive	site visit	7/18/2017	7/24/2017	8	4075.7699999999995	531	4606.7699999999995
557696451	7107	Iqra	Mcnally	Executive	supplier visit	6/21/2017	6/27/2017	21	4110.33	485	4595.33
127358183	3686	Kayan	Partridge	Purchasing	conference	2/24/2017	2/28/2017	11	4263.51	324	4587.51
477501922	5160	Amara	Graham	Executive	off-site meeting	2/13/2017	2/18/2017	16	4136.8099999999995	428	4564.81
180604167	3472	Macaulay	Delaney	Executive	site visit	5/11/2017	5/17/2017	21	4053.98	455	4508.98
477501922	4340	Amara	Graham	Executive	site visit	11/25/2017	11/30/2017	20	4018.7	470	4488.700000000001
180604167	6870	Macaulay	Delaney	Executive	supplier visit	12/15/2017	12/19/2017	17	3992.1000000000004	471	4463.1
948823811	6904	Steffan	Parker	Operations	off-site meeting	1/14/2017	1/19/2017	27	4193.01	266	4459.009999999999
103489296	4566	Ella-May	Jeffery	Sales	conference	7/27/2017	7/31/2017	18	4010.000000000005	320	4330
804757664	3510	Lubna	Flynn	Operations	off-site meeting	12/6/2017	12/10/2017	8	3958.07	360	4318.07
557696451	3854	Iqra	Mcnally	Executive	conference	3/7/2017	3/13/2017	24	3875.1099999999997	397	4272.11

Expense_DW 27 ×

Output

Query #4- This query calculates total travel spending by employee and includes their department. It identifies the highest-spending employees and shows how spending is distributed across departments. The finding is that Executives are the highest spenders followed by Sales people, the executive spending is quite a bit higher than the sales team.

```
SELECT
    Dept_name,
    Employee,
    First_name,
    Last_name,
    SUM(Total_Net_Amount) AS Employee_Total_Spending
FROM Expense_DW
GROUP BY
    Dept_name,
    Employee,
    First_name,
```

Last_name

ORDER BY Employee_Total_Spending DESC;

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

Dept_name	Employee	First_name	Last_name	Employee_Total_Spending
Executive	557696451	Iqra	Mcnally	54214.29
Executive	477501922	Amara	Graham	46766.159999999996
Executive	180604167	Macaulay	Delaney	40877.509999999995
Executive	303083671	Kit	Dunn	32210.420000000002
Sales	876723999	Precious	Robin	29553.749999999996
Sales	212757040	Oliver	George	29274.88000000001
Sales	587837897	Skye	Huang	28458.929999999993
Sales	703103749	Dillan	Yang	27937.499999999996
Sales	246605900	Vlad	Salgado	27659.630000000005
Sales	843647240	Lowen	Moran	26481.8
Sales	803094619	Finnian	Laing	26420.239999999994
Quality	575387889	Imaad	Bass	26335.200000000004
Quality	127808366	Rome	Mcdermott	26327.170000000002
Sales	267198729	Keri	Barrett	25825.4
Purchasing	836104663	Hilda	Reynolds	25465.98
Sales	832435710	Murtaza	Wheatley	25034.260000000006
Sales	830221763	Kashif	Cain	24995.050000000003
Sales	287632025	Marcie	Reader	24748.81
Sales	590463514	Korben	Goodman	24181.489999999998
Sales	758572377	Hawwa	Cassidy	23830.56
Sales	641790455	Ruairi	Yates	23805.600000000002
Sales	165558392	Leonidas	Porter	23530.69
Sales	103489296	Ella-May	Jeffery	23499.23
Executive	247505332	Morris	Chavez	23143.68
Sales	407413948	Shola	Ward	22931.25
Human Re...	225619358	Cindy	Hensley	22762.769999999997
Logistics	802792878	Mabel	Jimenez	22409.95

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